

Prediction, Discovery, and Explanation in Protein–Lipid Active Learning under Leakage-Controlled Evaluation

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Abstract

Acquiring strong binders and predicting binding affinity throughout a chemical space are distinct objectives. We study this distinction retrospectively using 864 dissociation-affinity and 549 inhibition-affinity protein–lipid pairs from BioDolphin, together with four exploratory target panels. An initial 5,850-run representation–kernel–policy benchmark is supplemented by 9,720 campaigns with cold-sequence and scaffold evaluation, frozen protein language-model features, and validation-only configuration selection. We implement prediction-aware Thompson sampling, an equal-rank combination of a joint posterior affinity draw and conditional predictive variance reduction. On cold-sequence dissociation data it reduces descriptive mean MAE from 1.541 to 1.406 pK relative to Thompson sampling, but pool top-decile recall falls from 59.6% to 23.3%. The result is a changed trade-off, not joint superiority. A further 1,296 campaigns evaluate small-corpus lipid self-supervision and initial-label-only feature adapters; improvements are inconsistent. Individual-bit nonlinear explanations, background sensitivity, fingerprint-collision audits, and illustrative contacts in two experimental complexes extend an earlier temporal SHAP analysis. Six-seed corrected tests are underpowered and establish no superiority. The study provides auditable methods and an interactive dashboard, while retaining limitations from assay heterogeneity, retrospective design, and limited independent biological data.

Keywords: active learning; protein–lipid affinity; Gaussian processes; Thompson sampling; protein language models; explainability

1. Introduction

Protein–lipid affinity prediction combines limited measured data, heterogeneous assays, and chemically diverse ligands. Active learning (AL) can make efficient use of sequential measurements, but efficiency depends on the objective. Selecting high-affinity pairs may enrich a measured collection while leaving poorly covered regions that dominate prediction error. Random-pair evaluation can further obscure generalization when related proteins and chemistries occur on both sides of a split.

We adapt the explainable ligand-affinity framework of [Srivastava et al. \(2026\)](#) to protein–lipid pairs. Our original random-pair grid found different configurations optimal for prediction and discovery. Here we turn that observation into an explicit dual-purpose selection rule and test it under harder splits. We do not claim that variance reduction, Thompson sampling, or their combination is a new theoretical principle. The contribution is a reproducible domain-specific implementation, matched evaluation, and careful accounting of where apparent improvements disappear.

The extended study contributes: (i) cold-sequence and scaffold evaluations with inner-validation model selection; (ii) frozen ESM-2 protein features crossed with five lipid encodings; (iii) prediction-aware Thompson and a pure variance-reduction ablation; (iv) leakage-controlled lipid adaptation; and (v) collision-aware and nonlinear explanation diagnostics,

including selected experimental contacts. The previous 5,850 campaigns remain a separate, descriptive discovery study. The new 11,016 unique campaigns do not turn that earlier post-hoc selection into independent validation.

2. Related Work and Scope

The ligand-affinity framework compares encodings, GP kernels, and scheduled exploration and exploitation, with explanations of acquired-model predictions (Srivastava et al., 2026). Gaussian processes provide joint uncertainty (Rasmussen and Williams, 2006); Thompson sampling queries a correlated posterior function realization (Russo et al., 2018). Predictive variance reduction is a long-established AL criterion (Cohn et al., 1996), and multiobjective batch selection has substantial prior art (Gupta et al., 2018). Accordingly, our selector is an empirical synthesis requiring comparison, not a novelty claim based only on a new name.

Frozen molecular encoders, including ChemBERTa and MoLFormer (DeepChem, n.d.; Ross et al., 2022), and protein encoders such as ESM-2 (ESM authors, n.d.), can improve representation without fitting every encoder on scarce affinity labels. They do not guarantee task-specific gains. We compare two ESM-2 sizes, not two independent architecture families. Domain continued pretraining and feature adapters are assessed separately from the frozen grid.

SHAP decomposes predictions relative to a background game (Lundberg and Lee, 2017). An ECFP position can represent multiple hashed environments (Rogers and Hahn, 2010); highlighting one occurrence is not an identification of a binding site. We distinguish numerical fidelity, stability, geometric contact evidence, and mechanistic validation.

3. Data and Evaluation Protocols

3.1. Published measurements and endpoint scope

BioDolphin is a broad lipid-protein interaction resource (Yang et al., 2024); only its curated quantitative affinity-bearing subset is used. Deduplication is by protein sequence and lipid SMILES, with curated median consensus labels and censored annotations excluded. Affinity is expressed as $pK = -\log_{10}(K/M)$; larger values indicate stronger binding. K_d and K_i are kept separate. Imported assay annotations can originate in other affinity resources; we do not claim BioDolphin generated all measurements. Its lipid definition is broader than membrane phospholipids.

The pooled endpoints contain 864 K_d and 549 K_i pairs. Across the pooled sets there are 549 lipids and 1,099 distinct protein sequences. TRAAK panels use the published mean first sequential binding constant from triplicates (Schrecke et al., 2021); replicates and later binding events are not independent training pairs. Two sequence-defined transthyretin subsets overlap pooled K_d . These panels are not six independent large biological datasets.

3.2. Original random-pair grid

Seeds 7, 19, and 42 define frozen approximately 20% test partitions. Pooled campaigns start with 60 labels and acquire ten batches of 30. Small panels start with two and acquire six more over ten phase slots, including zero-query slots. The design crosses six panels, five

Table 1: Distinct evaluation stages. Additional validation labels in the new stage are charged separately; final acquisition budgets alone are not total label cost. Tiny target panels appear only in the original exploratory grid.

Stage	Pairs	Seeds	Acquired labels	Selection/splits
Original pooled K_d	864	3	360	Post-hoc / random pair
Original pooled K_i	549	3	360	Post-hoc / random pair
TRAAK-a / TRAAK-b	16 / 16	3	8	Exploratory random pair
TTR-A / TTR-B	11 / 11	3	8	Exploratory random pair
New pooled K_d / K_i	864 / 549	6	120	Inner validation / three splits

lipid representations, five kernels, 13 policies, and three seeds, totaling 5,850 campaigns. Its reported winners were selected post-hoc from test summaries; these results are descriptive and not independent evidence of superiority. Test labels did not guide model fitting or within-campaign acquisition.

3.3. Cold-sequence, scaffold, and random controls

The new design uses seeds 7, 19, 42, 73, 101, and 137. Approximately 20% of whole groups are assigned to outer test, followed by 20% of the remainder to inner validation, subject to group sizes. The remaining acquisition pool starts with 24 labels and receives four batches of 24. Exact row indices and costs are recorded. Validation labels are never acquired, but are charged as additional labels. These 120-label campaigns are not budget-matched to the original 360-label study.

For cold-sequence evaluation, we compute all-pairs global Needleman–Wunsch alignments with BLOSUM62, gap-open 10 and extension 1. Identity is matched residues divided by alignment columns. Connected components join proteins with identity greater than 40% or the same recorded accession. Transitivity is retained; whole components are separated. All cold train/test splits have maximum global identity below 40%, with no exact sequence or recorded accession overlap. This operational definition does not establish curated family separation or exclude similar local domains.

Scaffold splits group nonchiral Bemis–Murcko scaffolds. All acyclic molecules share one group rather than being separated by their full SMILES. Random-pair splits provide a matched easier control. For each policy and partition, final inner-validation MAE selects representation and kernel, with deterministic lexical tie-breaking. Protein-specific comparisons select lipid/kernel while holding the protein representation fixed. This is repeated nested holdout with a single inner split, not full nested K -fold cross-validation. Outer test results never select configurations or candidates.

3.4. Metrics and inference

We report test MAE, R^2 , interval coverage, acquisition-pool top-decile recall, and label-normalized recall AUC. Recall includes initial and acquired pairs and is not a test classification score. Changing split regimes changes the eligible pool and group composition; differences are protocol sensitivity, not a controlled estimate of leakage alone.

Exploratory paired two-sided sign-flip tests use six seed-level differences, comparing prediction-aware Thompson with Thompson and uncertainty on MAE and recall AUC across six endpoint/split settings. Holm correction spans 24 tests. With six signs the smallest possible two-sided exact p is $2/64 = 0.03125$; the smallest attainable Holm-adjusted value in this family is 0.75. The analysis is therefore underpowered by construction. Overlapping holdouts are not independent biological replications; absence of significance does not establish equivalence. Future confirmatory work needs predeclared primary comparisons and independently justified sample size.

4. Models and Acquisition

4.1. Frozen representations and GP fitting

Lipid representations are radius-four, 4,096-bit ECFP, 167-bit MACCS, 384-dimensional ChemBERTa-MTR and ChemBERTa-MLM, and 768-dimensional MoLFormer-XL. Model revisions, feature hashes, and preprocessing are saved. MoLFormer removes stereochemistry as in its pretraining pipeline; representation comparisons therefore also reflect preprocessing assumptions. The original protein representation is a 26-dimensional composition/length/charge feature vector, not a learned structural encoding.

The new grid adds frozen ESM-2 8M and 35M features of dimensions 320 and 480. Mean residue pooling excludes special tokens. Five proteins longer than 1,000 residues are encoded in disjoint chunks and residue-weighted pooled; this avoids truncation but loses cross-chunk context. These encoders are not trained on the benchmark affinities. ESM-2 was pretrained externally, so benchmark sequence novelty relative to pretraining is not established.

The pairwise exact GP combines lipid covariance with a protein RBF factor when proteins vary. Initial-set statistics scale inputs and distances; only acquired labels determine target normalization. The original grid uses linear, Tanimoto, RBF, rational quadratic, and Matérn-3/2 lipid kernels. The new matched grid uses linear, RBF, and Matérn-3/2, spanning previous winner families without claiming an exhaustive kernel search. Constant mean, signal/noise variances, and applicable kernel parameters are fitted using analytic-gradient L-BFGS-B with warm starts and at most 60 iterations. New protein geometry is explicitly float64. A failed earlier float32 attempt is invalidated and excluded, not pooled with completed results.

4.2. Baselines and prediction-aware Thompson

The original suite contains uniform random, balanced UCB, five scheduled exploration and exploitation policies, joint Thompson, greedy mean, pure uncertainty, expected improvement, probability of improvement, and a fixed Thompson–diversity heuristic. EI/PI use marginal top- k scores, not joint batch EI. The new grid retains random, uncertainty, balanced UCB, and joint Thompson, adding the proposed implementation and pure variance reduction.

Let C be latent posterior covariance and τ^2 fitted observation noise in output units. A fixed proxy set V contains at most 64 inner-validation inputs, selected without labels. For

a candidate x , define

$$g(x) = \frac{1}{|V|} \sum_{v \in V} \frac{C(v, x)^2}{C(x, x) + \tau^2}. \quad (1)$$

This is the integrated reduction in proxy latent variance for one noisy observation with fixed GP hyperparameters, not expected reduction in MAE. Draw one correlated latent posterior function over the candidate pool, $\tilde{\mathbf{f}} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{C})$. Prediction-aware Thompson greedily scores unselected candidates by

$$s(x) = (1 - \lambda) \text{rank}_{[0,1]} \{\tilde{\mathbf{f}}(x)\} + \lambda \text{rank}_{[0,1]} \{g(x)\}, \quad \lambda = 0.5. \quad (2)$$

Ranks use average ties and are recomputed among remaining candidates. After each selection, condition the covariance using the noisy-observation Schur update before computing the next gain. The original function draw is retained within the batch. Pure variance reduction uses $\lambda = 1$; standard Thompson is separately implemented as the existing baseline. No weight search uses test outcomes. The selector receives proxy inputs but neither proxy labels nor outer-test inputs/labels. Covariance conditioning needs no fantasized labels; hyperparameters remain fixed within the batch.

The main grid is $2 \times 3 \times 6 \times 3 \times 5 \times 3 \times 6 = 9,720$ campaigns. All policies have the same initial rows, split, and budget within a partition. Formula checks match direct variance changes to 3.27×10^{-15} at worst. These checks establish implementation fidelity, not that variance is calibrated or that the chosen scalarization is optimal.

5. Lipid-Domain Adaptation

Label-free continued ChemBERTa MLM training begins from the pinned frozen checkpoint. Of 2,615 unique raw BioDolphin SMILES, seven fail RDKit validation. Excluding all benchmark scaffolds, including the acyclic group, and overlength token sequences leaves 790 molecules. This is a small pilot, not large-scale LMSD or SwissLipids pretraining. Three fixed epochs use 15% masking with the standard 80/10/10 masking rule, learning rate 10^{-5} , batch size 16, and seed 7. Embeddings are then frozen. Decreasing training loss is not independent evidence of better affinity representation.

The supervised alternative fits residual bottleneck feature adapters to ESM-2 35M and ChemBERTa-MLM embeddings using only each partition’s 24 initial labels. Bottleneck width is 32, residual scale 0.1, and final residual weights start at zero. A small predictive head optimizes standardized affinity MSE plus a 0.01 residual penalty with AdamW, learning rate 0.001, weight decay 0.01, and 100 fixed epochs. Adapted features are returned in the original feature units, avoiding a hidden scaling change. Each of 36 adapters is frozen and reused across policies/kernels before any acquisition. No future labels or full pooled-label fine-tuning are used. These are feature adapters, not attention LoRA or supervised transformer-weight updates.

The matched adaptation comparison fixes ESM-2 35M and the MLM lipid family, varying frozen, self-supervised, and adapter variants across all partitions, three kernels, and six policies. Its 1,944 entries include 648 frozen baselines already counted in the main grid, hence 1,296 additional campaigns. Kernel choice is by inner-validation MAE within each variant/policy.

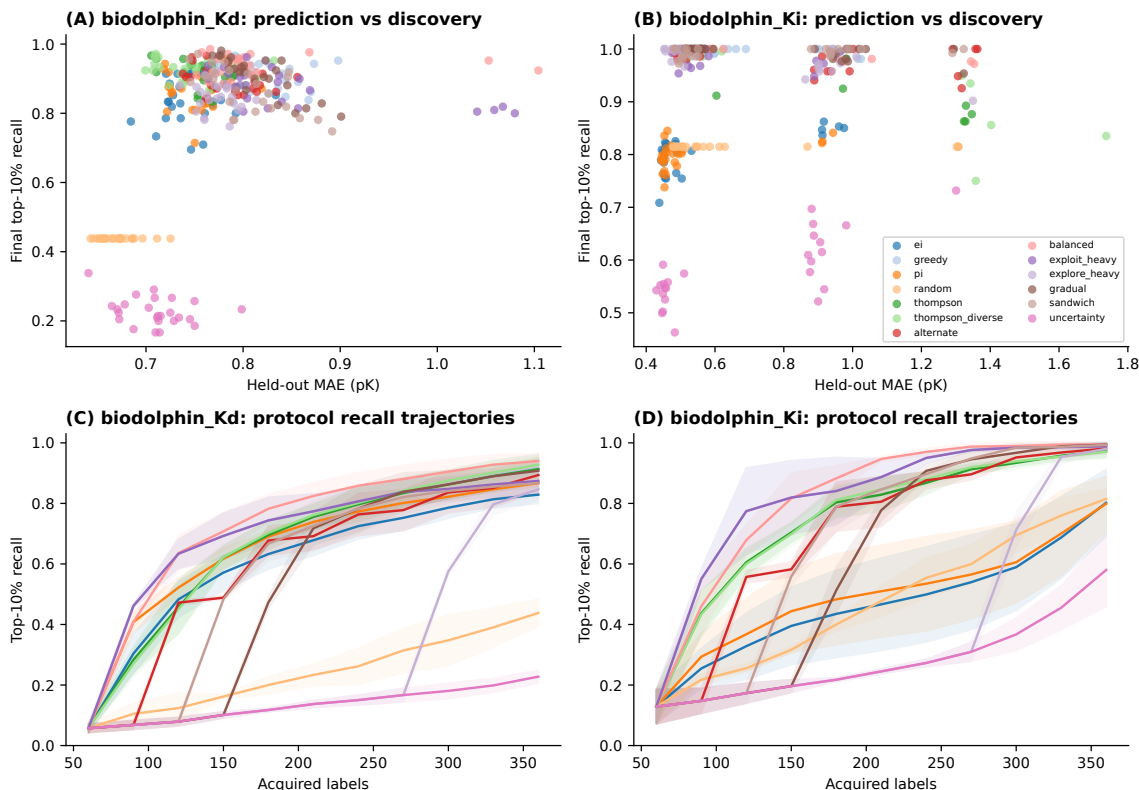


Figure 1: Original descriptive four-panel benchmark: prediction and discovery rank configurations differently. Pooled acquisition budgets are 360 labels; these curves are not budget-matched to the new 120-label study. Means across three seeds and post-hoc choices are exploratory.

6. Results

6.1. Original results and budget context

In the original post-hoc search, ECFP-linear uncertainty gives lowest descriptive pooled MAE: 0.641 ± 0.041 for K_d and 0.429 ± 0.044 for K_i . The lowest-MAE K_d Thompson configuration, MTR-Matérn, has MAE 0.705 ± 0.046 and recall 0.967. Balanced UCB with MLM-linear has the highest K_d recall AUC, 0.783, while greedy and exploit-heavy MTR-linear tie at 0.864 on K_i . These winners optimize different objectives. The K_i budget already covers about 82% of its acquisition pool; final recall alone can look favorable even without exceptional early discovery.

All tiny panels retain negative mean R^2 even for their lowest-MAE selected configurations. No reliable target-level prediction claim follows from 11–16 pairs. Full original tables and all historical figures remain in the supplement; their selections are not mixed with the validation-selected stage.

Table 2: Validation-selected six-seed means at 120 acquired labels plus validation cost. TS: Thompson; PA: prediction-aware Thompson. Recall is acquisition-pool top-decile recovery. No corrected superiority is established.

Endpoint	Split	TS MAE	PA MAE	TS recall	PA recall
K_d	Random	1.017	0.933	.513	.311
K_d	Cold sequence	1.541	1.406	.596	.233
K_d	Scaffold	1.305	1.277	.510	.264
K_i	Random	0.895	0.702	.643	.333
K_i	Cold sequence	1.419	1.428	.664	.319
K_i	Scaffold	1.469	1.336	.879	.340

6.2. Leakage-controlled evaluation exposes a harder task

Table 2 and Figure 2 show validation-selected six-seed outcomes. For prediction-aware Thompson, random-pair MAE is 0.933 on K_d and 0.702 on K_i , increasing to 1.406 and 1.428 on cold sequence, and 1.277 and 1.336 on scaffold splits. Random acquisition remains competitive: cold-sequence random MAE is 1.482 on K_d and 1.333 on K_i . Harder splits are essential to interpret the favorable original random-pair results.

Prediction-aware Thompson improves descriptive MAE over Thompson in five of six settings, but reduces recall in all six. On cold K_i , even MAE is slightly worse. Balanced UCB often has higher recall, whereas pure variance reduction frequently sacrifices discovery without achieving lowest MAE. Thus adding variance reduction does not resolve the trade-off; it provides an explicit operating point. No calibrated clinical, experimental-cost, or economic utility has been specified to justify this operating point as best.

6.3. Protein and lipid representation gains are mixed

Figure 3 holds prediction-aware Thompson fixed and selects lipid/kernel separately within each protein representation. Cold K_i MAE is 1.446 for composition, 1.411 for ESM-2 8M, and 1.368 for ESM-2 35M. Cold K_d values are 1.471, 1.482, and 1.451. On scaffold K_d , the 8M encoder is descriptively best among these three (1.223 versus 1.295 composition and 1.259 for 35M). Model size therefore does not produce a monotonic improvement. These conditional comparisons differ from allowing inner validation to select the protein representation too.

The adaptation study also has mixed outcomes (Figure 4). For prediction-aware Thompson on cold K_d , frozen / adapter / SSL MAE is 1.431 / 1.416 / 1.520. Cold K_i gives 1.567 / 1.692 / 1.562. Scaffold K_d improves descriptively to 1.252 with adapters and 1.219 with SSL, from 1.295 frozen. These comparisons fix the MLM lipid family and should not be confused with the full five-representation selected results. There is no separate confirmatory adaptation significance claim.

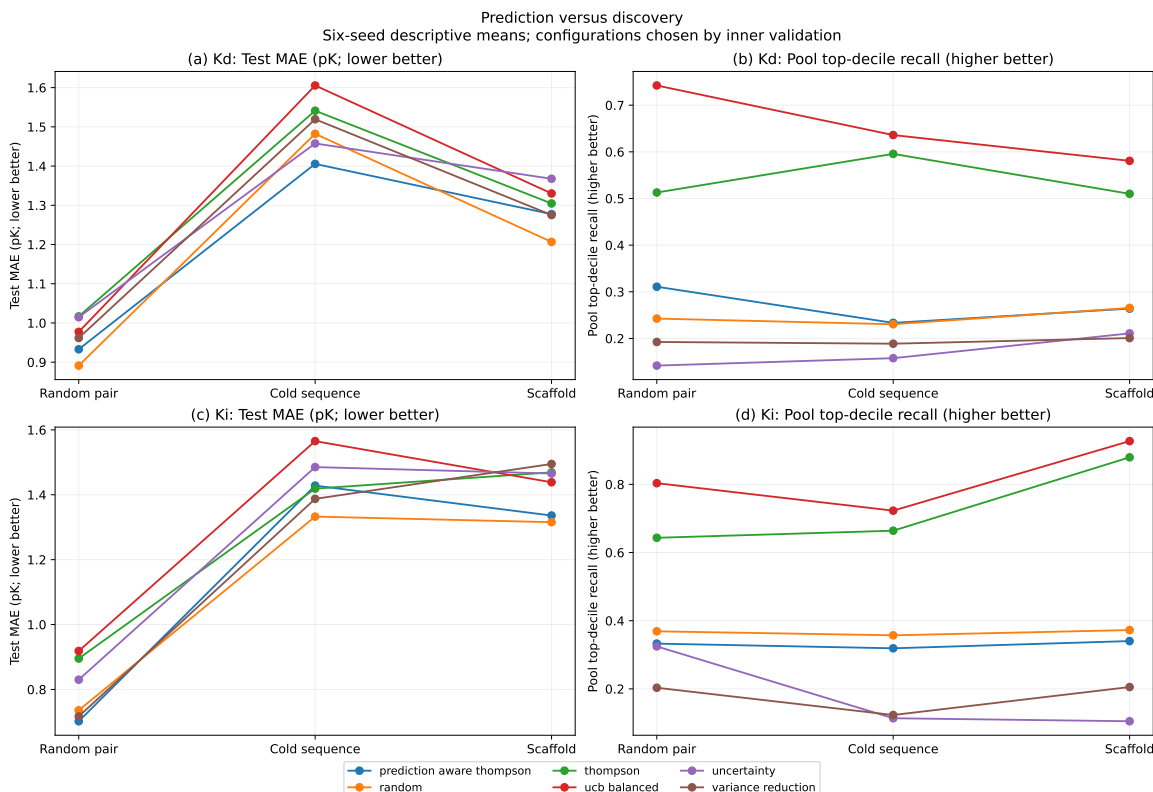


Figure 2: New six-policy study in four panels: outer-test MAE and pool recall for K_d and K_i . Each point is a six-seed mean after inner-validation configuration selection. Lines connect split regimes for readability, not time; no uncertainty bands or significance claims are implied. Raw seed values and SD summaries accompany the paper.

7. Explanation Analysis and Structural Checks

7.1. Grouped and temporal explanations

Historical grouped SHAP reports cover 54 selected final models. The temporal analysis covers 36 ECFP-linear campaigns, 396 snapshots, and 14,754 query-model explanations with the query protein fixed. Affine conditional predictions admit exact individual-bit contributions; independent KernelExplainer comparisons agree below 10^{-10} pK. This establishes computational fidelity for the specified background game, not uniqueness of the explanation or causal importance.

Backgrounds and unqueried populations can change across cycles. Importance trajectories therefore combine changes in the model and explained population; they are not pure longitudinal biological signals. Affinity-prioritized fragment display uses measured labels retrospectively, including held-out rows. Those labels do not enter SHAP, querying, or fitting but make the illustrations unsuitable as independent validation.

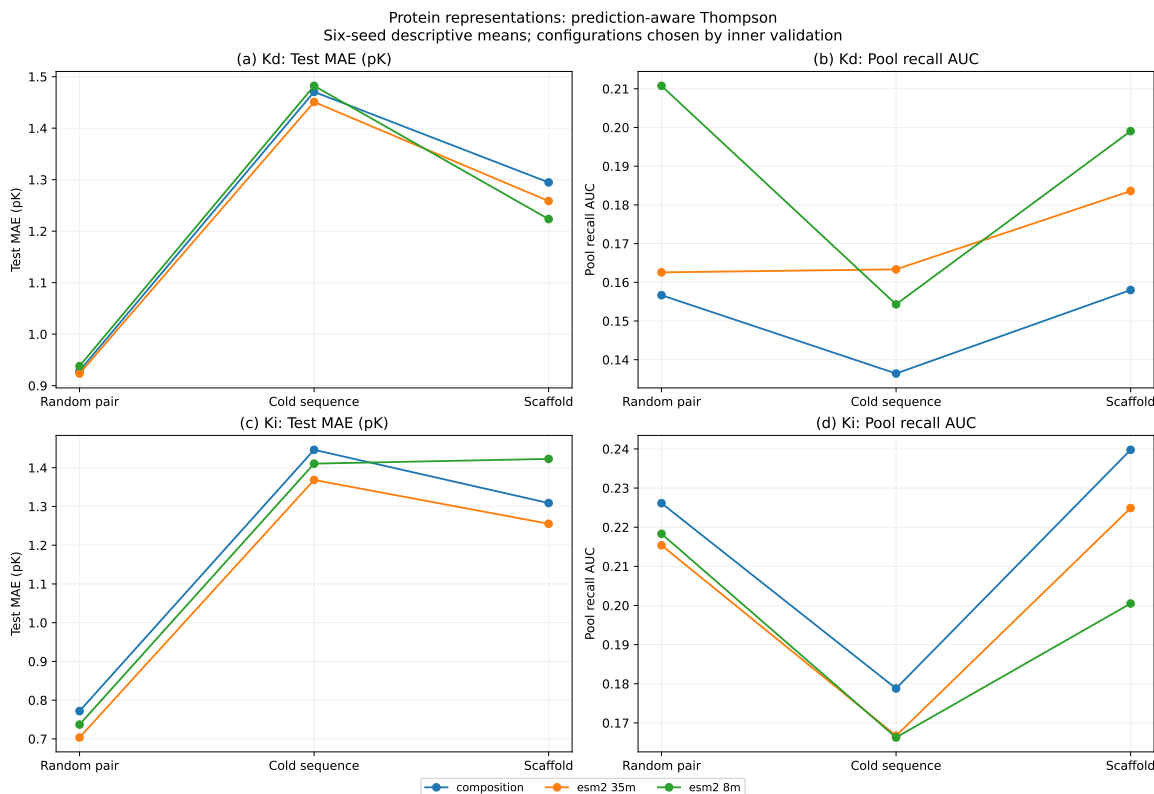


Figure 3: Four-panel protein comparison for prediction-aware Thompson. Lipid representation and kernel are chosen by inner validation within each protein group. The two ESM-2 sizes are one architecture family; differences include checkpoint pre-training and dimension, not only a controlled capacity change.

7.2. Hash collisions and nonlinear background sensitivity

Across the 549 lipids, 3,539 of 3,980 observed ECFP positions contain more than one distinct unfolded Morgan identifier. Folding sets are verified against the actual fingerprint. Even a 32-bit unfolded identifier is not proof of unique chemistry, and fragment strings have contextual limitations. High/low-affinity collision comparisons use all labels retrospectively and are not independently validated structure–activity relationships.

We additionally estimate individual-bit permutation Shapley values for four prespecified historical nonlinear models: K_d/K_i crossed with RBF and Matérn, ECFP, Thompson, seed 7. Two fixed test queries per model, three eight-pair acquired-training backgrounds, and two repeats yield 48 explanations. Each estimate uses eight antithetic permutations; protein features are fixed. Dummy bits are exactly zero. Coalition predictions are evaluated from exact GP sufficient statistics; attribution estimation itself is approximate. Additivity does not prove convergence.

Mean top-ten Jaccard across backgrounds is 0.698 for RBF and 0.721 for Matérn, versus 0.970 and 1.000 for same-background repeats. In this small diagnostic set, background

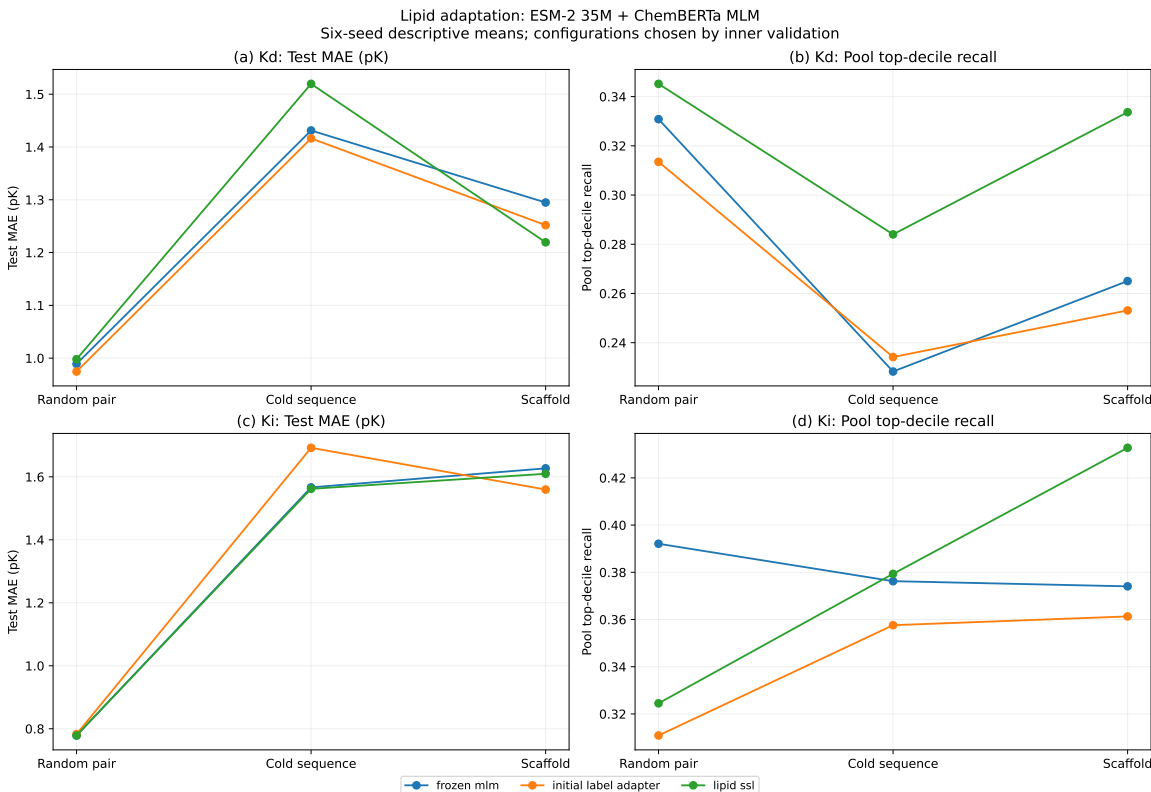


Figure 4: Four-panel matched adaptation comparison using ESM-2 35M, ChemBERTa-MLM family features, and prediction-aware Thompson. Only kernels are selected by inner validation. Initial-label adapters and a 790-molecule SSL pilot do not yield consistent gains; 648 frozen baselines are reused, not counted as additional unique experiments.

choice affects ranking more than repeat sampling. The few models and queries do not support a broad robustness guarantee.

7.3. Experimental contact illustrations

We match four affinity-prioritized historical fragment examples to experimental complexes 4OAS and 6CDJ using recorded BioDolphin source identifiers, chains, residues, and ligand CCD codes. There are only two distinct complexes, not four independent confirmations. Full heavy-atom graph matches are used, with 24 and 16 symmetry mappings respectively; no absent atoms are invented. Mapping ignores chirality. A contact is a heavy-atom distance at most 4 Å to the recorded protein chain in the first PDB model, without biological-assembly expansion.

All atoms in these small highlighted fragments contact the protein under all enumerated mappings. The 4OAS examples contact HIS96, LYS94, and VAL93; 6CDJ examples contact GLY48 or GLY48/49. About 54% and 51% of all ligand atoms, respectively, also

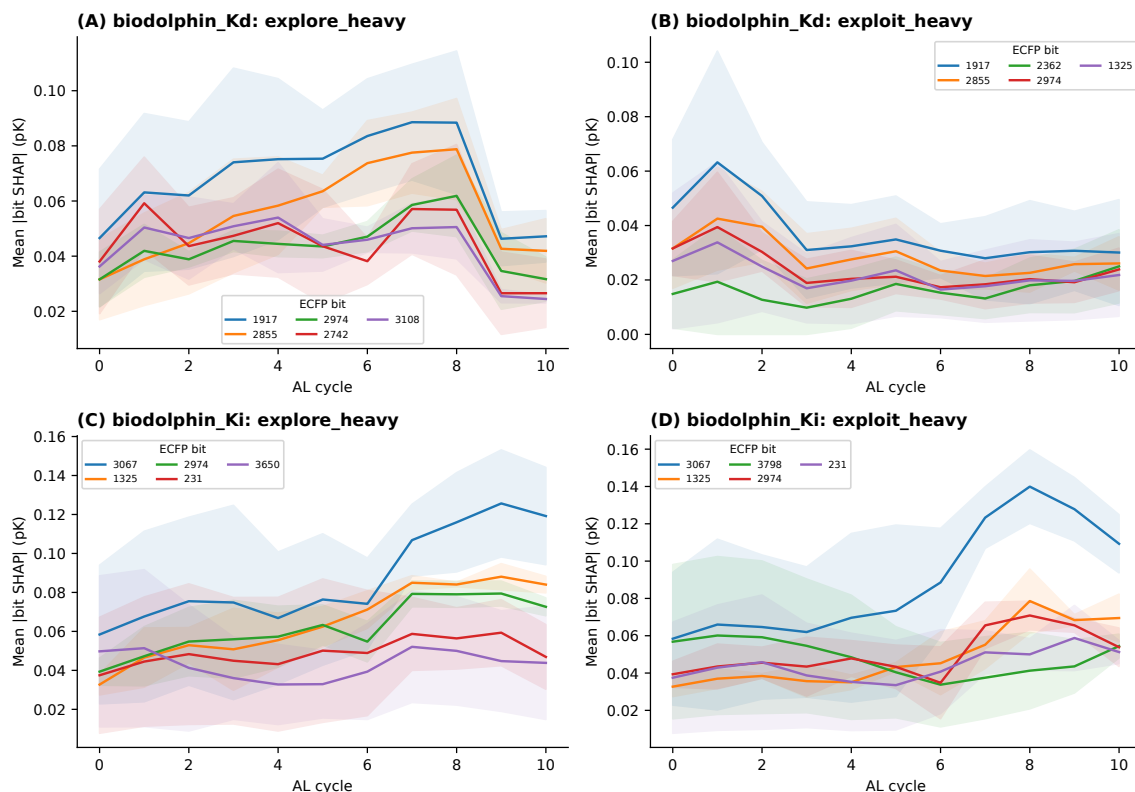


Figure 5: Four historical ECFP-linear temporal panels. The five bits with largest final seed-averaged importance are shown; bands are seed SD. Explore-heavy and exploit-heavy schedules are contrasted. Changing background/query populations prevents a causal interpretation of trajectory changes. These are the original 360-label campaigns.

satisfy the cutoff. Selected-contact agreement is therefore illustrative, not proof that high SHAP uniquely localizes a binding site. No docking, binding-energy decomposition, residue attribution, prospective mutagenesis, or global contact-map validation is claimed.

8. Dashboard, Reproducibility, and Limitations

The dashboard separates the original complete grid, grouped SHAP, highlighted structures, and the new study explorer. The latter exposes all 9,720 main campaigns and 1,944 adaptation comparisons through study, endpoint, split, protein, lipid, adaptation, kernel, policy, and seed dropdowns. Metrics and CSV downloads are derived from saved results; browsing does not fit models. Plots based on inner-validation selection are distinguished from raw configuration browsing to discourage test-score selection.

The accompanying source repository is available at <https://github.com/harshitsinghsnu/LipidAL>. A Vercel-ready static export is pro-

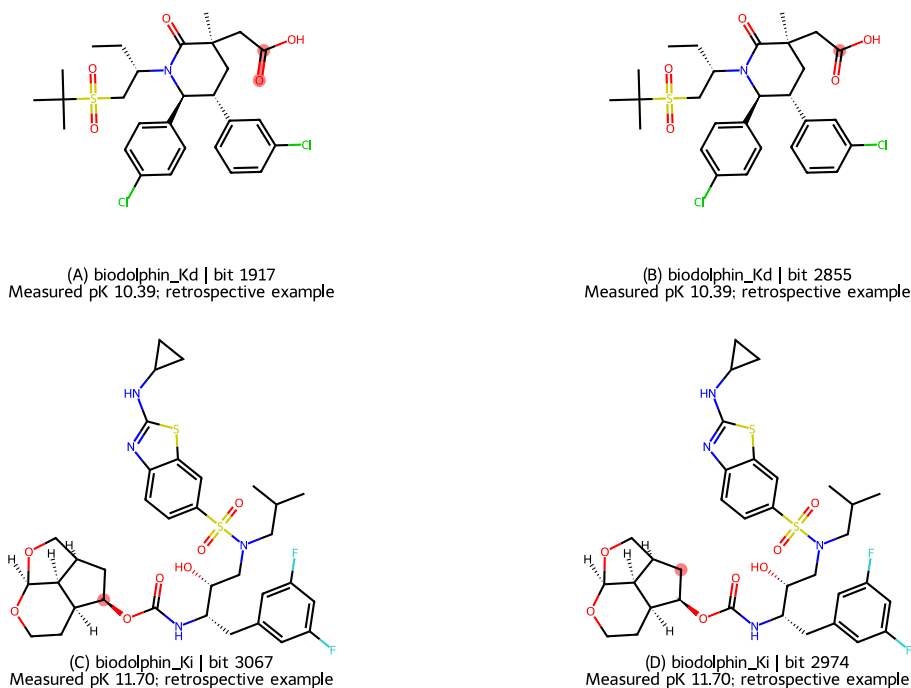


Figure 6: Four historical affinity-prioritized fragment panels, subsequently checked against two experimental complexes. Highlighting shows a recorded ECFP environment, not a unique motif or a newly discovered binding site. The same molecule may appear for different bits. Geometric contact checks are reported separately and do not make these selected examples causal.

vided; a public deployment URL has not yet been verified for this manuscript build. Repository ownership identifies the author, so these links must be anonymized or removed for double-blind review.

The public export contains allowlisted derived tables, figures, explanations, and the manuscript. It excludes source datasets, model weights, full GP checkpoints, private uploaded data, and local environments. Uploading measured data and running campaigns remain features of the local Python application. A public AL service would additionally require authenticated uploads, data-retention rules, persistent job storage, a queue, and bounded workers; the static site does not claim to provide these.

The original audit records 5,850 campaigns and 48,750 fits, with 26 fits not meeting the optimizer criterion. The new main audit covers 9,720 campaigns, 48,600 fits, and 23 nonconverged fits. Adaptation checks cover 1,944 comparisons, 36 initial-only adapters, zero benchmark-scaffold overlap in SSL training, and six nonconverged fits including reused baselines. Initial sets, validation separation, proxy-label exclusion, selection rules, and

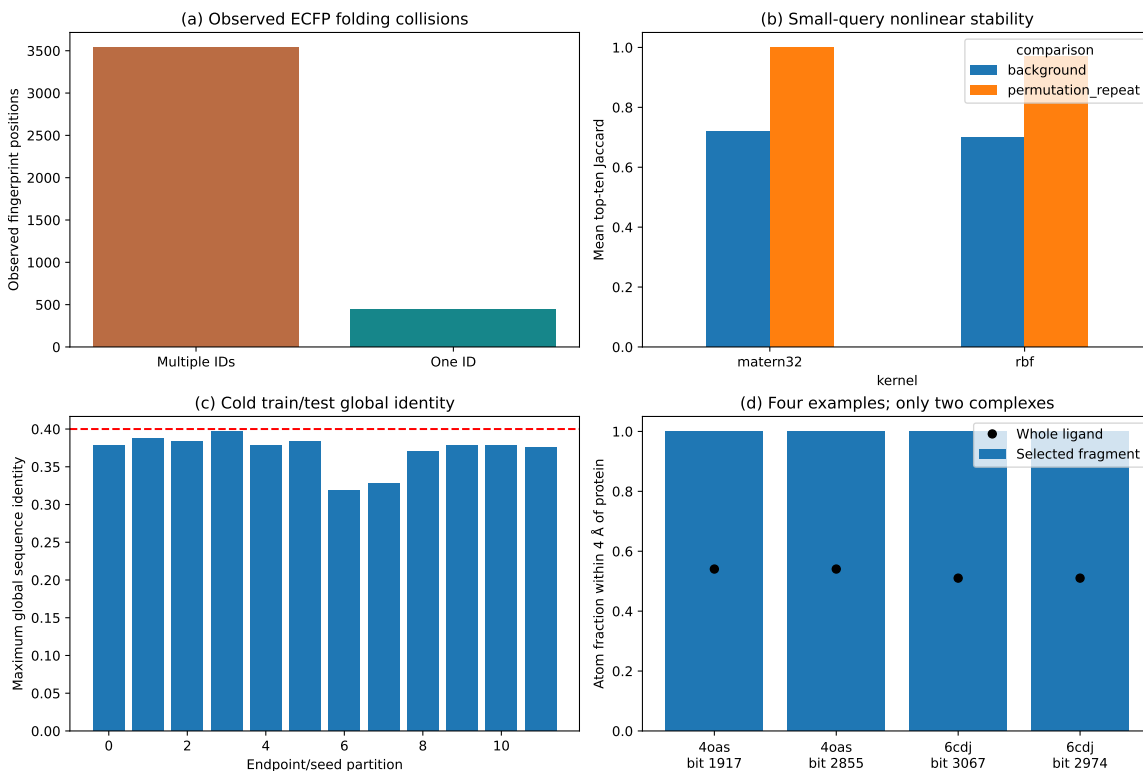


Figure 7: Four diagnostics: (a) multiple unfolded identifiers per observed ECFP position; (b) background versus sampling-repeat top-ten stability in the limited nonlinear query set; (c) maximum cold train/test global sequence identity; (d) selected-fragment and whole-ligand contact fractions for four examples from only two complexes. These checks do not establish unique motifs, biological causality, or comprehensive explanation robustness.

data/source hashes are checked. Convergence flags are retained rather than silently discarded.

Assay heterogeneity remains unresolved. Consensus labels can merge different conditions; recorded ranges and source annotations are not a full replicate noise model. Zero observed range does not imply noiseless affinity. Split control does not remove dataset selection bias, related local domains, pretraining overlap, or all chemistry similarity. Large cold groups can change endpoint and chemical distributions. The new work reuses datasets seen during earlier development, so it is not external validation.

Statistical and biological scope are limited: the tests are underpowered, the adaptation corpus is small, only one protein-model family is compared, and contact checks cover selected examples. No prospective measurement or time-stamped future holdout has been fabricated. Supervised pooled-label fine-tuning before splitting would leak future labels into AL and is deliberately not performed. Larger independent corpora, assay-aware mod-

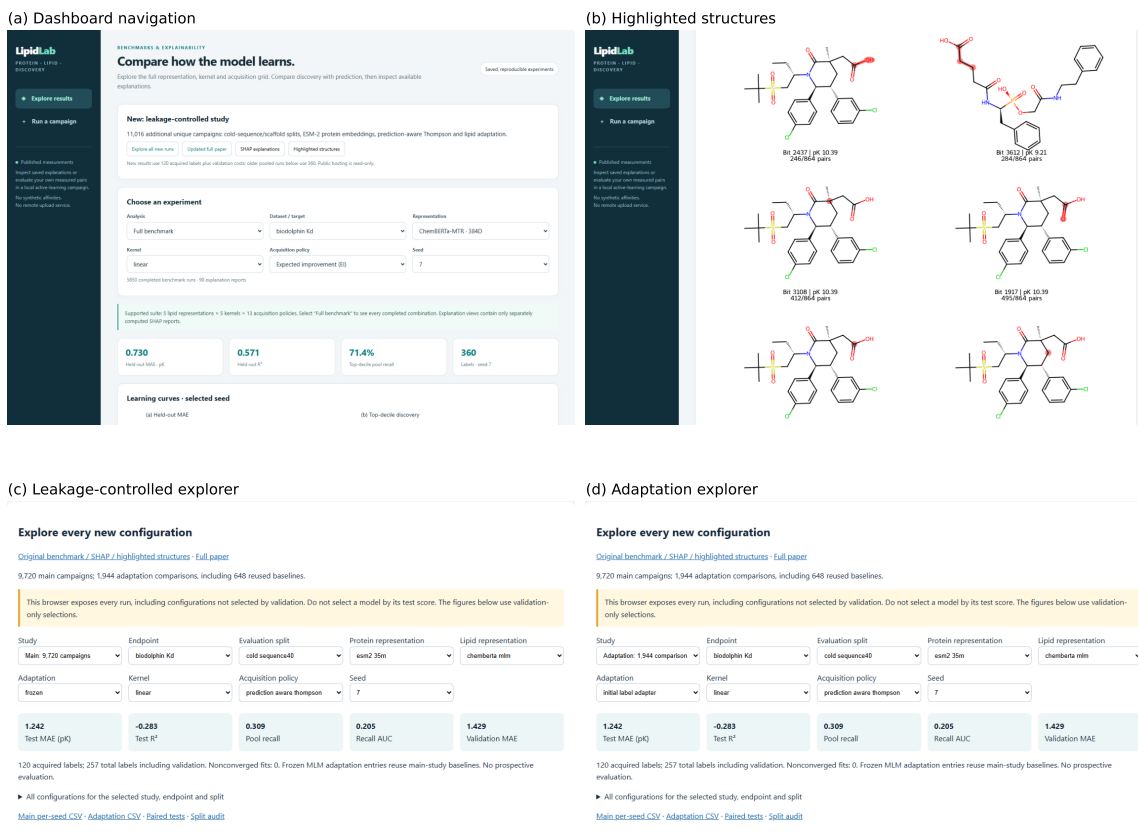


Figure 8: Four genuine screenshots of the updated static dashboard: original-grid navigation, highlighted fragment view, leakage-controlled configuration selection, and adaptation selection. Captures use a local HTTP server serving the generated read-only dashboard, not mockups or evidence of a public deployment. Hosting status is stated separately in the text.

eling, distinct protein encoder families, and adequately powered prospective studies remain necessary before stronger claims.

9. Conclusion

Protein–lipid AL has distinct predictive and discovery objectives. Explicit prediction-aware Thompson changes their balance but does not dominate standard policies. Cold-sequence and scaffold evaluation reveal substantially harder prediction than random pairs. Frozen protein embeddings and small lipid adaptation offer mixed gains, while nonlinear/collision/contact diagnostics make explanations more auditable without making them mechanistic. The principal outcome is a transparent empirical framework and a qualified trade-off, not established algorithmic novelty or prospective superiority.

Supplementary Material

The source package includes complete per-seed and summary tables for both evaluation stages, adaptation comparisons, paired tests, split and execution audits, explanation diagnostics, and hashes. The original atlas retains all 223 historical source figures in four/five-panel composites; an updated composite supplement includes all seven original main figures, the three new four-panel comparisons, and the updated dashboard. The main paper uses a subset to avoid redundant figures. Original data/checkpoints are unchanged. The adopted ACML template and page budget do not imply current submission eligibility; author details, anonymity, source redistribution, and scientific claims require author review before submission.

References

- David A. Cohn, Zoubin Ghahramani, and Michael I. Jordan. Active learning with statistical models. *Journal of Artificial Intelligence Research*, 4:129–145, 1996. doi: [10.1613/jair.295](https://doi.org/10.1613/jair.295).
- DeepChem. ChemBERTa-77M MTR and MLM checkpoints. <https://huggingface.co/DeepChem/ChemBERTa-77M-MTR>; <https://huggingface.co/DeepChem/ChemBERTa-77M-MLM>.
- ESM authors. ESM-2 pretrained models and implementation. <https://github.com/facebookresearch/esm>. Pinned checkpoint revisions and pooling details accompany the feature manifest.
- Sunil Gupta, Alistair Shilton, Santu Rana, and Svetha Venkatesh. Exploiting Strategy-Space Diversity for Batch Bayesian Optimization. Proceedings of AISTATS, PMLR 84:538–547, 2018. <https://proceedings.mlr.press/v84/gupta18a.html>.
- Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30, 2017. <https://arxiv.org/abs/1705.07874>.
- Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006. <https://gaussianprocess.org/gpml/>.
- David Rogers and Mathew Hahn. Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling*, 50(5):742–754, 2010. doi: [10.1021/ci100050t](https://doi.org/10.1021/ci100050t).
- Jerret Ross, Brian Belgodere, Vijil Chenthamarakshan, Inkit Padhi, Youssef Mroueh, and Payel Das. Large-scale chemical language representations capture molecular structure and properties. *Nature Machine Intelligence*, 4:1256–1264, 2022. doi: [10.1038/s42256-022-00580-7](https://doi.org/10.1038/s42256-022-00580-7).
- Daniel Russo, Benjamin Van Roy, Abbas Kazerouni, Ian Osband, and Zheng Wen. A tutorial on Thompson sampling. *Foundations and Trends in Machine Learning*, 11(1):1–96, 2018. doi: [10.1561/22000000070](https://doi.org/10.1561/22000000070).

Samantha Schrecke, Yun Zhu, Jacob W. McCabe, Mariah Bartz, Charles Packianathan, Minglei Zhao, Ming Zhou, David Russell, and Arthur Laganowsky. Selective regulation of human TRAAK channels by biologically active phospholipids. *Nature Chemical Biology*, 17:89–95, 2021. doi: [10.1038/s41589-020-00659-5](https://doi.org/10.1038/s41589-020-00659-5).

Satya Pratik Srivastava, Rohan Gorantla, Sharath Krishna Chundru, Claire J. R. Winkelman, Antonia S. J. S. Mey, and Rajeev Kumar Singh. Explainable active learning framework for ligand binding affinity prediction. *Digital Discovery*, 5:769–779, 2026. doi: [10.1039/D5DD00436E](https://doi.org/10.1039/D5DD00436E).

Li-Yen Yang, Kaike Ping, and Yunan Luo. BioDolphin as a comprehensive database of lipid–protein binding interactions. *Communications Chemistry*, 7:288, 2024. doi: [10.1038/s42004-024-01384-z](https://doi.org/10.1038/s42004-024-01384-z).